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The Economics of Food Supply Chain Resilience

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Abstract

Food supply chain resilience has become a priority for policymakers in recent years. Prompted by several systemic disruptions and the increased likelihood of future shocks, significant attention and public financial investment have been devoted to preparing supply chains to be more able to absorb shocks and more nimbly respond in the future. Food and agricultural supply chains have been particularly scrutinized due to widespread shortages and food price inflation of key staples in recent years. Varied perspectives have emerged about the factors that contribute to the resiliency of food supply chains and the interventions that will be most effective. This article reviews the perspectives on food supply chain resilience generally. Furthermore, we attempt to synthesize the methodologies that have been used to study resilience and offer some guidance for economists studying the topic moving forward.

1. INTRODUCTION

Food supply chains experienced a multitude of disruptions in recent years, beginning with the COVID-19 pandemic. These events sparked a global dialog about the resilience of our food systems. While many critiqued the facets of existing food production and processing in response to these shocks, others applauded food supply chains' ability to adapt to unprecedented disruptions. Nevertheless, the increasingly probable nature of these types of events, whether due to natural disasters, extreme weather, geopolitical events, or disease (including those in plants and animals), likely means that the resilience of supply chains, especially those of food and agriculture, will remain a prominent public concern.

The US federal government and many state governments have heavily invested in making food supply chains more resilient. But despite ongoing investment, there exists a considerable debate about what constitutes resilience and the factors that support resilient supply chains. Thus, it is unclear whether these policies will effectively meet their targets, and what trade-offs, if any, will result from these interventions. As these conversations continue, there is opportunity for agricultural economists to advance supply chain resilience as a field of research and have a voice in these dialogs.

The concept of supply chain resilience is by no means new. The disciplines of operations research and supply chain management have long focused on structuring businesses in privately optimal ways by balancing business-as-usual conditions with investments for future uncertainty. However, for economists, supply chain resilience has primarily garnered interest in the wake of the COVID-19 pandemic, perhaps driven by the public policy discourse. As Just (2023) highlights, agricultural economists are well positioned to speak to the vulnerabilities of supply chains and discuss the potential impacts of disruptive events. As a discipline, it is imperative that we synthesize the lessons learned from the pandemic and the subsequent impacts, including the responses taken by policymakers. Moving beyond the pandemic, there remain significant opportunities to advance theoretical and empirical tools to assess resilience, counterfactual scenarios, and any potential costs (e.g., efficiency trade-offs and public expenditures) of interventions.

This review aims to provide an overview of the perspectives about supply chain resilience, including the dense literature since the COVID-19 pandemic, and offer considerations for economists studying these questions as we move beyond the lingering impacts of COVID-19. The disruptions these recent events caused (and may cause in the future) no doubt affected all sectors of the economy. Therefore, the lessons discussed here may be applicable to all sectors of the economy, but we give a special focus to agricultural and food supply chains given that shoring up the food supply to these shocks has been a public policy priority.

The review is organized as follows. In Section 2 we discuss food supply chain vulnerabilities, including sources of vulnerability, lessons from the COVID-19 pandemic, and the unique characteristics of agrifood supply chains. Section 3 examines perspectives on resilience across three broad strands of literature: food system resilience (including social-ecological-systems and household perspectives), firm-level resilience, and sectoral resilience. The section concludes with a discussion of policy considerations in addressing resilience. In Section 4, we turn to defining, modeling, and measuring resilience in economics, including theoretical considerations and empirical methods. Section 5 offers concluding remarks, including identifying research gaps and opportunities.

2. FOOD SUPPLY CHAIN VULNERABILITIES

Agriculture and food supply chains are particularly vulnerable to shocks from a variety of sources. This has been brought to the forefront of public concern by a series of shocks that have impacted global food supply chains in recent years, including the COVID-19 pandemic, climate and

environmental shocks, disruptions caused by animal and plant diseases, and geopolitical events, such as the Russia–Ukraine war. The nature of food production is such that markets may be uniquely compromised when macro shocks do occur. These factors include, but are not limited to, the perishability of products, the short-run inability to adjust supply, labor-intensive production processes, and production and manufacturing being increasingly concentrated geographically and among a few firms.

2.1. Sources of Vulnerability

Major geopolitical events such as the Russia–Ukraine war can upend global commodity markets, disrupting trade flows and global supply chains. Supplies of key inputs (natural gas, oil, fertilizer) and major agricultural staples such as wheat and oilseeds have been severely disrupted by the recent conflict, triggered by the Russian invasion of Ukraine in February 2022. Prior to the conflict, Russia and Ukraine accounted for 12% of total calories traded (Glauber et al. 2023). The Black Sea region (Ukraine and Russia) represented more than a quarter of global wheat exports in 2021, while Ukraine and Russia together accounted for around 55% of the global supply of sunflower oil and almost 20% of the global corn market (Ferguson & Ubilava 2022, Dyson et al. 2023). The disruption to global food supply chains was exacerbated by other countries imposing export restrictions on food as a response to turmoil in global commodity markets.¹

Animal and plant diseases, along with foodborne pathogens, can disrupt food supply chains, albeit the impacts are usually sector specific. Outbreaks of highly pathogenic avian influenza (H5N1) in wild birds and poultry disrupted poultry and egg supply chains across the United States and Europe in 2022, leading to shortages of eggs. *Escherichia coli* contamination in romaine lettuce caused upheaval in North American fresh produce markets in late 2018–2019, with major economic losses for shippers, processors, and retailers (Spalding et al. 2023). In the case of foodborne pathogens, typically these events are (relatively) short-lived and may result in switching behaviors among consumers where substitute products are available. Effective traceability systems that enable the source of contamination to be identified relatively quickly help reduce the scope, scale, and economic burden of foodborne diseases. Animal and plant diseases that become endemic to a region or across regions pose longer-term challenges to agricultural production and the vulnerability of food supply chains.

Agricultural and food systems are exposed to a variety of environmental forces, from slow-moving changes in average conditions, to larger anomalous disruptions such as floods, droughts, extreme heat, hurricanes, pests, and diseases. It is anticipated that the production of key staples such as maize, wheat, livestock, fish, and seafood will continue to be negatively impacted by rising global temperatures and changes in precipitation (Davis et al. 2021, IPCC 2022).

Extreme weather events create both localized and global shocks (Fan et al. 2021). Regions with poor access to global supply chains are vulnerable to localized extreme weather events. Nevertheless, many shocks are both global and cross-sectoral in nature (Baldwin & Freeman 2022). Davis et al. (2021) point to the 2008 food crisis, where drought in key grain-producing regions, exacerbated by rising demand for biofuels, high energy prices, decreasing grain stocks, and a depreciation of the US dollar, led to a spike in global grain prices. High prices triggered a series of ill-conceived policy responses, such as rice export bans, further exacerbating disruption in global

¹This happened quickly. As Glauber et al. (2023) note, the number of countries imposing export restrictions on food in response to the Russia–Ukraine conflict rose from 3 to 16 within six weeks of the onset of the conflict. Five (widely traded) agricultural commodities accounted for the vast majority of export restrictions: wheat, palm oil, corn, sunflower oil, and soybean oil. By October 2022, 21 countries had implemented export bans (Dyson et al. 2023).

supply chains and worsening the food crisis. Baldwin & Freeman (2022) argue that in normal times global supply chain shocks may be geographically and/or sectorally concentrated in terms of direct impact, citing the example of the 2011 Great East Japan earthquake and tsunami and the ensuing Fukushima nuclear accident. However, they note that more recent events (including the COVID-19 pandemic) have generated broader shocks that are global and cross-sectoral in nature.

2.2. Lessons from COVID-19

While interest in supply chain resilience is not new (see, e.g., Stone & Rahimifard 2018), the COVID-19 pandemic catapulted supply chain resilience into the public consciousness and onto the agendas of policymakers. From initial shortages in personal protective equipment and key medical devices, to longer-run challenges in sourcing key components (e.g., microchips in automotive manufacturing), the pandemic caused a myriad of supply chain disruptions. Perceived challenges to resilience in food supply chains, however, are a particularly sensitive area garnering significant attention.

In reflecting on the implications of the COVID-19 pandemic for food supply chain resilience, we need to disentangle short-run effects from longer-run structural issues, as well as consider the nature of the shocks to supply chains. The pandemic, and the measures taken to contain the spread of the virus and protect human health (lockdowns, movement restrictions, etc.), resulted in short-run disruptions to food supply chains caused by supply and demand imbalances and infrastructure constraints. A large literature has emerged examining the implications of the COVID-19 pandemic for agrifood supply chains, largely finding that developed country food supply chains rebounded relatively quickly from the initial short-run shocks during the early stages of the pandemic (see, e.g., Hobbs 2021, 2022; Orden 2021; Ramsey et al. 2021).

The COVID-19 global pandemic was unprecedented in scope and scale. The correlated nature of the shocks from the global pandemic, affecting all stages of multiple supply chains simultaneously, is noted by Hadachek et al. (2024). Short-run retail demand shocks arose from the sudden switching of household food expenditures from food service to food retail, exacerbated by consumers' short-run stockpiling behaviors. Simultaneously, upstream supply disruptions occurred due to labor shortages and processing plant closures, particularly in the fresh produce and meat sectors.

Vulnerabilities emerged in some sectors, while others remained relatively robust. The food service sector (and, by extension, firms supplying food service outlets) was significantly affected by lockdown measures and the sudden shift in household food expenditures from food service to food retail (Chenarides et al. 2021, Goddard 2021, Malone et al. 2021, de Souza et al. 2022). The meat processing sector proved vulnerable to labor-force outbreaks of COVID-19 (Luckstead et al. 2021, Bina et al. 2022, Wahdat & Lusk 2023). Subsequent increases in food prices in the later periods of the pandemic are associated with a recovery in demand (for goods in general) following the easing of lockdown measures, along with logistical bottlenecks (Vos et al. 2022a).

In a developing country context, the short-term direct and indirect spillover effects of lockdown measures on supply chains, household incomes, and poverty are noted by Vos et al. (2022b). Large-scale modern, integrated food supply chains proved more robust than traditional supply chains. Resilience stemmed from a greater capacity to innovate and the ability to switch suppliers and destination markets due to a stronger resource basis and a more extensive supplier network (Vos et al. 2022b). The vulnerability of traditional domestic supply chains in developing countries to pandemic-induced disruptions relative to export-oriented modern supply chains is a recurring theme in the literature (see, e.g., Van Hoyweghen et al. 2021).

It is beyond the scope of this article to discuss in detail the effects of the COVID-19 pandemic on food systems, which is the subject of a large emerging literature (for reviews, see

Béné 2020, Hobbs 2022, Vos et al. 2022a). Nevertheless, three lessons emerge from the pandemic: (a) widespread, correlated shocks have the potential to cause more disruption relative to isolated, idiosyncratic shocks; (b) commodity sectors and supply chain segments differ in their vulnerability due to physical attributes (storability versus perishability), reliance on labor (degree of mechanization), and existence of alternative sources of supply; and (c) we should distinguish between short-run disequilibrium, where imbalances in supply and demand (entirely expected) cause price spreads to diverge in vertically linked markets, and medium- to longer-run disruptions that are structural in nature.

2.3. What Makes Agrifood Supply Chains Unique?

Agrifood supply chains differ from manufactured goods supply chains in several important respects, with implications for resilience. Agricultural production is dependent on biological processes creating stickiness in supply: Crops and livestock take time to mature, and once production is underway, it cannot be halted or delayed. Agriculture is also subject to regular, and sometimes catastrophic, weather shocks within the growing season, creating market volatility that cannot fully be anticipated at the time production decisions are made. While some commodities (e.g., grains) are storable, seasonality and perishability characterize many agricultural commodities, reducing the ability to alter delivery schedules in response to a disruption (Hobbs 2021). Globally traded commodities (e.g., wheat, pulses, oilseeds) are reliant on well-functioning global transportation networks, including ocean freight.

Market power and anticompetitive considerations have long been a focus of attention in agrifood supply chains. Unlike in the manufacturing goods sector, many agricultural supply chains are characterized by large numbers of relatively small producers facing concentrated input supply markets (e.g., fertilizer, seed, agro-chemicals) along with both buyer and seller power in the food processing and retailing stages of production. The efficiency and distributional implications of market power and closer vertical coordination in agricultural markets are nuanced, given economies of scale and product differentiation advantages arising from larger-scale, more closely coordinated supply chains (Sexton 2013). The structure of agrifood supply chains, and the nature of vertical relationships within that supply chain, should factor into economic assessments of food supply chain resilience (see Section 4.1).

Insights from the broader resilience literature suggest that investing in excess capacity in imperfectly competitive markets provides a buffer from exogenous shocks, bolstering resilience (Gnutzmann et al. 2020). Within supply networks for complex manufactured goods, firm-level strategies to build robustness and mitigate fragility include investing in redundancy through large inventories and multisourcing (Elliott & Golub 2022, Elliott et al. 2022). One could argue that agrifood supply chains featuring commodities that are (at a broad level) generic, substitutable, and less complex provide inherent robustness relative to complex manufactured goods. The relatively short-lived duration of market disruptions in key agricultural commodities initially at the outset of the COVID-19 pandemic (relative to complex manufactured goods sectors) may reflect this difference. Designing effective policies to enhance food supply chain resilience depends crucially on the what, how, and why of resilience: understanding what we mean by resilience, how we define and measure resilience, and the intended outcomes from enhanced resilience.

3. PERSPECTIVES ON RESILIENCE

Interest in resilience spans three broad perspectives: food system, firm-level, and sectoral. Much of the focus on food system resilience arises in the development and climate change literatures, with an emphasis on food security and resilience among vulnerable households, and the resilience of the

Table 1 Perspectives on resilience: summary of key concepts

Perspective	Definition	Core concepts	Measures
Food systems: social-ecological	The ability of a social or ecological system to absorb disturbances while retaining the same basic structure and ways of functioning, the capacity for self-organization, and the capacity to adapt to stress and change (IPCC 2007)	Stability Robustness Vulnerability Adaptive capacity	Ecological indicators/indexes Biodiversity Steady states
Food systems: household and food security	Capacities of households and communities to cope with shocks in ways that are not detrimental to their long-term well-being (Béné 2020)	Absorptive capacity Adaptive capacity Transformative capacity	Self-assessment Nutritional and caloric values Psychometric techniques
Firm level	Ability of a firm and its supply chain to return to their former states following a severe disruption	Inherent resilience (intrinsic and anticipatory) Adaptive resilience Static resilience (existing resources) Dynamic resilience (new investments)	Time-to-recover Time-to-survive Risk exposure index Cost-effectiveness of resilience strategies (benefit-cost ratio)
Sector level	The ability of a sector to adjust to disequilibrium/return to equilibrium prices/quantities following a shock, while minimizing welfare losses to producers and consumers	Efficiency versus flexibility Economies of scale Risk and uncertainty Input substitutability Disequilibrium Counterfactuals	Equilibrium outcomes Distributional impacts Welfare measures Cost-efficiency trade-offs

food system as a whole to exogenous shocks. Supply chain resilience from a firm-level perspective emphasizes physical product flows, logistics, and intraorganizational management systems. Following the COVID-19 pandemic, there is increased interest in understanding sectoral-level resilience (e.g., in meat processing) or resilience within a spatial context, such as the debate over shorter, localized supply chains versus diversified, global supply chains. Embedded within these three broad perspectives on resilience are a variety of disciplinary perspectives, including ecology, sociology, engineering, logistics, management science, and economics. A sampling of literature across these perspectives reveals different definitions of resilience depending on the intended outcome, along with common underlying themes. We summarize the key concepts from the different perspectives in resilience in **Table 1**.

3.1. Food Systems: Social-Ecological-Systems Perspective

From a systems perspective, the notion of resilience has its roots in ecology and the study of resilience and stability in ecological systems (such as predator and prey relationships) (Folke 2006). Holling (1973, p. 17) proposed resilience as the capacity (of a system) to persist in the face of change, arguing that resilience “determines the persistence of relationships within a system and is a measure of the ability of these systems to absorb changes of state variables, driving variables, and parameters, and still persist.” Building upon this approach, the concept of social resilience focuses on the ability of human communities to withstand and adapt to exogenous shocks (Folke 2006). Much of the interest in resilience in the context of climate change and food security adopts the social-ecological-systems perspective, examining the dynamic relationship between humans

and the environment and society's capacity for change. In this vein, the Intergovernmental Panel on Climate Change (IPCC) defines resilience as: "The ability of a social or ecological system to absorb disturbances while retaining the same basic structure and ways of functioning, the capacity for self-organization, and the capacity to adapt to stress and change" (IPCC 2007, p. 86). Within the social-ecological-systems literature, therefore, resilience refers to "a system's ability to either withstand, absorb, or appropriately respond to disruptions" (Coopmans et al. 2021, p. 2).

Measuring the resilience of social-ecological systems is challenged by the difficulty in defining and operationalizing appropriate metrics. Cabell & Oelofse (2012) suggest that agroecosystem resilience is evaluated in three ways: (a) the amount of change a system can withstand, (b) the ability of a system to respond through self-organization, and (c) its capacity for learning and adaptation. Resilience implies the ability for adaptation and change. A system that is resilient today may not be resilient in the future if it fails to adapt to external shocks. Paradoxically, resilience in the short run may reduce a system's resilience in the longer term if it engenders complacency or resistance to change (Cabell & Oelofse 2012).

Common concepts across the food systems resilience literature include stability, robustness, vulnerability, and adaptability. Urruty et al. (2016) draw a distinction between stability, robustness, vulnerability, and resilience in the context of agricultural systems. In their view, stability refers to the constancy of agricultural outputs over time or spatially, such as crop yield stability. Robustness refers to the ability to maintain desired levels of agricultural outputs despite weather, price, and disease disruptions, implying flexibility. Vulnerability refers to the degree to which agricultural systems are likely to be harmed by disruptions, such as risk, market failures, and extreme weather events. Finally, resilience is described as the ability to absorb change and anticipate future disruptions through adaptive capacity (Urruty et al. 2016). Nevertheless, operationalization of these four concepts remains challenging because they cannot be observed directly but must be inferred using indicators and proxy measures.

3.2. Food Systems: Household and Food Security Perspective

In the context of food security considerations, resilience refers to the capacities of households and communities to cope with shocks in ways that are not detrimental to their long-term well-being (Béné 2020). Shocks may be covariate, affecting numerous individuals simultaneously (e.g., drought or floods in a specific region), or idiosyncratic and specific to a household (e.g., illness) (Ansah et al. 2019). Literature in this sphere focuses on resilience at the household level. Food security is achieved when a household has stable access to sufficient, healthy, and nutritious food. The extent to which households can cope with and adapt to disruptive shocks is a signal of resilience. Thus, a resilient food system maintains food security in the face of shocks (Ansah et al. 2019).

In the context of smallholder households, Fan et al. (2021) frame food system resilience as resilience of consumption, resilience of production, and resilience of storage, marketing, and trade. Resilience results from a set of capacities or abilities (social, human, financial) that allow a household (or economic agent) to withstand a shock without resorting to negative coping strategies (such as selling off livestock or other productive assets) (Béné 2020). Whether resilience is achieved depends on the extent to which a household or agent can utilize these capacities to bounce back from a shock. As Béné (2020) notes, resilience capacities are inputs to the resilience process, while resilience per se should be viewed as an intermediate outcome, rather than the ultimate goal. In this vein, studies have attempted to quantify resilience capacities, identify resilience-building strategies, and develop a set of outcome indicators.

Three commonly used indicators for food system resilience capacity include absorptive capacity, adaptive capacity, and transformative capacity. Absorptive capacity refers to the ability of a system (or household) to withstand shocks and recover, for example, altering the timing of harvesting to avoid floods or delaying debt repayments. Adaptive capacity connotes the ability to build flexibility to alternative scenarios, such as the use of drought-resistant crop varieties. Transformative capacity refers to the ability of systems and institutions to adapt over the medium to longer term (Ansah et al. 2019). As an inherently latent variable, the same challenges apply in quantifying resilience. Typical approaches include the use of proxy measures, indicators, and self-assessment using psychometric techniques. We discuss empirical challenges in Section 4.2.

3.3. Firm-Level Perspective

A firm-level perspective on supply chain resilience originated in the supply chain management, operations research, and logistics literatures, with an emphasis on the ability of a system to return to its former state following a severe disruption. Rooted in engineering, this approach to resilience emphasizes physical product flows and the achievement of system deliverables. For example, Gao et al. (2019) identify three relevant metrics: time-to-recover, time-to-survive, and the risk-exposure index. Time-to-survive is the period after a shock during which the firm's supply chain is still capable of serving normal demands, while time-to-recover is the time between a disruption and when the supply chain returns to full functionality. The risk-exposure index quantifies a firm's risk of total lost sales under disruptions. Inventory management and protection are a core focus of this approach to resilience. The emphasis on inventory as a primary risk mitigation strategy distinguishes the manufactured goods sector from many agrifood supply chains characterized by perishability and stickiness of supply.

In their review, Stone & Rahimifard (2018, p. 214) compare an engineering definition of resilience ("the ability of a system to return to an equilibrium or steady state after a disturbance") with an ecological definition ("the magnitude of the disturbance that can be observed before the system changes its structure"). They note that many studies of resilience in the supply chain management literature adopt an adaptive definition of resilience, which focuses on adaptive change to exogenous shocks rather than states of equilibrium. In this vein, Ponomarov & Holcomb (2009, p. 131) describe firm-level resilience as: "The adaptive capacity of the supply chain to prepare for unexpected events, respond to disruptions, and recover from them by maintaining continuity of operations at the desired level of connectedness and control over structure and function."

A strand of literature in production economics examines the cost-effectiveness of investing in resilience, modeling firm-level decisions with respect to the optimal combination of inputs given the risk of input disruptions (see, e.g., Dormady et al. 2019, 2022). This literature distinguishes between inherent versus adaptive resilience and static versus dynamic resilience. Inherent resilience reflects capacities already built into a system or which can be easily repurposed. Inherent resilience includes intrinsic resilience, such as excess capacity or the degree of input substitutability, and anticipatory resilience, such as stockpiling essential materials or equipment. Adaptive resilience is associated with improving the efficiency of input usage in the production process, for example, technological improvements that enhance the substitutability of inputs. Static versus dynamic resilience refers to the extent to which efficient use of existing resources can be used to mitigate the effect of disruptions (static) versus new investments or the setting aside of resources to facilitate reestablishment of production at a future date (dynamic) (Dormady et al. 2022).

Dormady et al. (2022) identify 11 firm-level resilience tactics, including input substitution, resource isolation (managing without a critical input), holding inventories, investing in excess capacity, relocating production, resource pooling, conservation (reducing input usage), production recapture, import substitution, technological change, and improving management effectiveness.

The cost-effectiveness approach to the economics of resilience utilizes three theoretical concepts: avoided losses, maximum potential losses, and implementation cost. Avoided losses are the extent to which sales revenue or output would have been lower in the absence of a given resilience tactic. Maximum potential losses are the degree of losses a firm would encounter in the absence of any explicit actions to enhance resilience. Implementation cost is the cost of implementing a tactic or group of tactics (Dormady et al. 2022).

Building upon these three concepts, two operational resilience metrics are defined: a benefit-cost ratio for a given resilience tactic (avoided losses divided by tactic cost) and the resilience metric (avoided losses divided by maximum potential losses). Using firm-level survey data, Dormady et al. (2022) apply these metrics to an analysis of firm-level resilience tactics following two extreme weather events in the United States: Superstorm Sandy in 2012 and Hurricane Harvey in 2017.

Combining insights from the food systems and firm-level resilience literatures, Kabir et al. (2023) examine resilience in fish and shrimp value chains in Bangladesh during the COVID-19 pandemic. Using firm-level survey data, they examine whether factors identified as promoters of enterprise-level resilience—diversity, scope and scale of activities, extent of vertical integration, and involvement with clusters—are positively associated with resilience indicators including business continuity, revenues, household food security, and self-assessed resilience measures related to the perceived ability of the firm to respond to challenges, diversify, and overcome disruptions.

Nevertheless, the multidimensional nature of resilience, and the challenges in operationalizing the concept, makes it difficult to assess how a firm is performing; for example, did firm survival come at the expense of profitability? As with the food systems perspective, defining the intended outcome is important: Is it resilience per se or some other measure of firm success? Challenges also exist in disentangling the drivers of resilience when individual firms are resilient across one dimension but vulnerable across others.

3.4. Sectoral Resilience Perspective

Between the big picture of food system resilience, and the microfocus on resilience at the firm level, lies industry or sector level resilience. Much of the interest in agrifood supply chain resilience in the wake of the COVID-19 pandemic focused on challenges (and successes) in specific agrifood sectors, such as meat processing, fresh produce, and food retailing. Unlike firm-level analysis, where the agent is the private firm engaging in restructuring its supply chain to optimize performance, at the industry or sector level, the agent is typically government/policymakers with the broader goal of maximizing social welfare (Azzam 2023).

How resilience is defined and what resilience goals are remain elusive questions. The perceived lack of resilience of the US meat packing sector during the COVID-19 pandemic ignited fierce policy debates over whether and how to address resilience in the sector. In a theoretical contribution, Azzam (2023) shows that if the goal of resilience is to ensure farmers have uninterrupted access to more packing plants, simply increasing the number of small packing plants is unlikely to achieve that objective, except under strict conditions. Hadachek et al. (2024) find that the policy of stimulating market entry in meat processing is most effective in supply chains with high market power, given gains in consumer and producer surplus, but considerably more limited where industries are already highly competitive.

The pandemic highlighted vulnerabilities in labor-intensive industries, such as meat processing. As Bina et al. (2022) note, the primary disruption was to operational capacity rather than physical capacity. In this case, policy interventions aimed at increasing physical capacity (e.g., facilitating the entry of more small-scale processing plants) are unlikely to enhance supply chain resilience. Measured in terms of cattle slaughter volumes, empirical analysis shows that US regions more heavily dependent on large slaughter plants were generally not more affected by

pandemic-related (labor) shutdowns. The economies of scale and scope advantages of larger processing plants and the potential trade-offs between efficiency and flexibility are frequently noted, pointing to the need to have a very clear understanding of the desired outcomes of policy interventions that seek to enhance resilience (Hobbs 2021, Lusk et al. 2021, Ma & Lusk 2021, Cooper et al. 2023).

The vulnerability of US food manufacturing industries to disruptions in input supply sectors is evaluated by Wahdat & Lusk (2023). Using an input-output framework, vulnerability of a sector is characterized by the diversity of intermediary input purchases (e.g., by the meat processing sector or the grain handling sector). The degree of diversification in input supply is evaluated using a diversification index. Knowing the degree of input-purchase diversification can help identify food sectors that are prone to cascade effects from exogenous shocks. Nevertheless, it is unclear how input diversity could be increased at a sectoral level; for example, the animal processing sector, by its nature, is dependent upon inputs of cattle, hogs, and poultry from the farming/ranching sector.

While much of the focus on supply chain resilience examines vertical supply chain relationships (producers-processors-retailers-consumers), the role of lateral supply chains (logistics, labor, materials) is often overlooked. Reardon et al. (2022) refer to these elements as the blood and bones of food systems, where bones refers to transportation networks and infrastructure and blood to logistics services and regulatory governance. Gupta et al. (2023) examine the relationship between logistics quality and performance and the intensity of food market disruptions across Indian states during the COVID-19 pandemic. The analysis points to specific institutions that are needed to build resilient supply chains, including the quality of logistics services, the efficiency of regulatory processes and facilitation, the timeliness of cargo delivery, and the availability of logistics infrastructure. Resilience in this context is evaluated at a household level using household food disruption data as an indicator of resilience, as well as at a market level using wholesale market arrivals. States with poorer-quality logistics services were associated with higher levels of household food disruption (lower food consumption expenditures), while wholesale market arrivals recovered more quickly from lockdown in states with stronger logistics services.

3.5. Strategies to Enhance Resilience

Although defining the alternative definitions of resilience is straightforward, what can and what ought to be done to enhance resilience are more nuanced questions. As discussed in Section 3.3, firms optimize and manage risk in their supply chains according to their private risk preferences and expected profits. However, it is unlikely that the private level of risk of the firm matches the socially optimal level of risk to society more generally (Grossman et al. 2021, Baldwin & Freeman 2022). The wedge between the privately optimal level of risk and publicly optimal level reflects social inefficiency, whereby welfare losses to consumers could be mitigated by more resilient production and processing. Herein lies the justification of policy intervention to correct a market failure and promote more resilient supply chains. Inefficiencies of this type may be stark in agri-food supply chains given social and health externalities of food insecurity that are unlikely to be accounted for by private firms (Just 2023). Whether or not governments can achieve (and through what manner) outcomes closer to the socially optimal levels of risk in the agrifood sector remains an open theoretical and empirical question.

As Baldwin & Freeman (2022) note, information asymmetries (about collective risks) may result in suboptimal decisions by individual firms. If information about complex, interrelated supply chain risks has public good properties, the private sector will underinvest in the collection of this information. The complexity of supply chain relationships makes it difficult for firms to have full information about their entire value chain across multiple inputs and the ancillary services that

support a myriad of supply chain activities. Widespread correlated shocks exacerbate this challenge. Lack of full information may result in firms underestimating risks and hence suboptimal investments in resilience tactics. Resolving these information problems may require public sector efforts to improve the assessment of systemic risks, identifying points of vulnerability, and facilitating access to firm-level data to allow a more transparent assessment of risks.

Firms may differ in their capacities to invest in flexibility to enhance supply chain resilience. Market structure considerations are relevant here. Firm-level strategies to enhance resilience may be more likely to emerge in larger firms with economies of scale and scope advantages. Alternatively, the absence of competitive pressure in concentrated markets may weaken the incentive for investments in resilience tactics. Understanding how market structure affects the incentives of private actors to invest in resilience is an empirical question, yet it remains an important consideration in designing effective policy interventions to enhance resilience.

Strategies to enhance resilience also entail policies to reduce barriers to global supply chains and requires that countries commit to a rules-based international trading system rather than reverting to distortionary measures in times of crisis (Hertel et al. 2021). Identifying and addressing cross-border logistics bottlenecks may also require policy intervention if the incentives of private actors do not align with broader social welfare goals.

Food is a necessity, such that threats (real or perceived) to food supply chain resilience are an emotive, politically sensitive topic. Policy responses walk a fine line between bolstering resilience in the short run and introducing unintended consequences that exacerbate market disruptions and inefficiencies in the medium to long run. Ferguson & Ubilava (2022) note that nearly three-quarters of emerging market and developing economies took policy actions attempting to moderate the effect of the 2008 food price crisis. Export restrictions contributed further to the rise in global commodity prices, a pattern that was repeated during the recent COVID-19 pandemic and during the Russia–Ukraine conflict. Popular measures attempting to enhance the resiliency of domestic food supply chains, but with the potential for unintended consequences, include renewed focus on competition laws, regulating perceived price gouging, facilitating the entry of new processing firms, and promoting the geographical diversification of food systems into new regions (Hadachek et al. 2024). Implementing policy measures without accounting for the competitive conditions within a market, or the efficiency implications of smaller-scale and/or more geographically dispersed production systems, may do more harm than good.

Resilience is a multidimensional concept, examined through various disciplinary lenses and at different levels of analysis (food system, household, firm, sector). Commonalities include flexibility, diversity, and the capacity to adapt, respond, and rebound. Less clear is what the intended outcomes of resilience should be, rather than viewing resilience as an end in and of itself. Questions of resilience (whether it be at the food system level, individual firm level, or supply chain level) are inherently questions of economics: supply and demand shocks, costs of production and input substitutability, risk and uncertainty, market efficiency, and welfare outcomes. We turn now to a consideration of defining and measuring resilience in economics.

4. MEASURING RESILIENCE IN ECONOMICS

To date, discussions of food supply chain resilience in economics have largely been qualitative. One challenge, as discussed in the previous section, is the lack of congruency in the definition of resilience across disciplines, and even within economics itself. Amid these varied perspectives, there remain considerable differences about the measures of resilience, the factors that make supply chains more resilient, and the impacts of policies that aim to achieve supply chain resilience. Here, we review some of the recent methodological approaches in this literature and provide some guidance for future work.

4.1. Theoretical Insights and Considerations

Historically, modeling agricultural and food supply chains in economics has largely used static, risk-free models. For example, Gardner (1975) generates partial equilibrium predictions for retail-farm price spreads from shifts in supply and demand in a competitive market.² The outcomes of interest in models of this type are almost exclusively average market outcomes, such as price, quantity, and welfare measures. As we have discussed, however, the question of resilience concerns dynamic decisions in the face of idiosyncratic shocks, including the extremes of the distribution. To that end, traditional static models of behaviors within the supply chain cannot fully capture the dynamic objectives and decisions involved for agents, where decisions today impact their ability to absorb shocks in the future.

As discussed in Section 3, the perspective the modeler is attempting to capture can lead to very different conclusions. For example, the resilience of individual firms in food processing does not necessarily imply that the overall system or sector is resilient and vice versa. Just as Tack & Yu (2021) highlight about defining risk, differing perspectives may result in competing or contradicting effects for farmers, intermediaries, and consumers. Hence, when discussing resilience as a concept and in practice, one must first carefully identify what type of resilience (e.g., household, firm, sector) is being discussed and the definition of resilience within that context.

From the perspective of the firm, modeling the economic incentives is somewhat straightforward: a profit-maximizing firm that faces uncertainty over a set time horizon. The resilience of a given firm, then, can be defined by several variables: time-to-recover, revenue volatility, labor outcomes, or other indicators of firm performance. Studying firm-level resilience, however, is a primary focus of the separate field of supply chain management and operations research.³ In theory, firms should internalize the private value of resilience and make decisions accordingly, but the firm's objective function would ignore the value of resilience to the general public.

One recent example of firm-level resilience analysis is the use of a real options framework in Chenarides et al. (2021). They present a stylized example, where onion producers have significant financial incentive to maintain flexible marketing channels if one channel is shut off. Producers then must weigh that value with the associated switching costs or the costs to maintain those channels during normal periods. However, private risk management practices by firms are costly and reap infrequent and uncertain rewards. For firms in many food sectors the cost-inefficiencies of worst-case-scenario planning may often exceed the private expected benefits. Hence, competition and economies of scale from the most cost-efficient firms during normal periods may drive out the firms and strategies best suited to survive extreme periods. Thus, socially optimal levels of food supply chain resilience may not arise organically.

Herein lies the justification for public policy intervention. But recurring government expenses to enhance or subsidize supply chain resilience similarly may be too great to implement. Ultimately, the optimal point on the resilience-efficiency trade-off curve depends on the size of cost-inefficiencies and the magnitude of welfare impacts from infrequent extreme events. Quantifying and valuing market-wide welfare ramifications seem to represent the area where economists have the comparative advantage. While methodologically challenging, credibly valuing different

²As agriculture and food markets became more industrialized, models of food supply chains began to incorporate imperfect competition as most modern agricultural markets deviate from perfectly competitive conditions to some extent (Holloway 1991).

³See Simchi-Levi et al. (2014) for a descriptive application of these concepts. As previously stated, supply chain resilience and risk management of the firm are by no means new concepts or new methodologies. Rather than provide a detailed overview of this discipline, we present a few recent examples from the economics literature.

forms of protection from historically unprecedented events and the costs of doing so would provide fruitful insights for the economics discipline and public policy.

Recent evaluations of the food supply chain have made some progress in this regard. Wahdat & Lusk (2023) reveal that labor-dependent segments of the food supply chain, such as animal processing, are most vulnerable to future disruptions. They demonstrate this point by constructing diversification indexes for upstream inputs and comparing exposure across food manufacturing sectors.

Azzam (2023) develops a homogeneous product, competitive-fringe model to test whether expanding the processing capacity of the fringe, which represent small- and medium-sized beef processors, impacts the overall capacity of the processing sector amid shocks. He predicts that the strategy by the US government will only work where the fringe and incumbent share the market equally and compete in price, but this is unlikely to hold in practice.

Hadachek et al. (2024) construct a static, oligopoly-oligopsony model of a generic food supply chain and use Monte Carlo simulations to evaluate how large perturbations impact both mean welfare outcomes and their relative variance across a number of policy counterfactual scenarios. The findings indicate that market power in intermediary processing is detrimental to food supply chain resilience. Complementary to Azzam (2023), the resilience gains primarily result from the price competition of entrants, rather than the additional capacity itself. Other interventions, such as anti-price gouging laws, stimulating processor entry, and geographic diversification, have somewhat mixed impacts on supply chain resilience and often come at the cost of economic efficiency.

As economic work proceeds in this arena, three considerations stand out. First, while many perspectives on resilience exist, economists seem best positioned to address the sector-wide resilience of food systems. This is arguably the least-explored dimension of resilience but is critical to society more broadly. Second, any model that attempts to capture the resilience of agrifood supply chains must also carefully capture the cost-efficiency trade-offs associated with strategies or policy interventions. Third, there remains little understanding of the public benefit of more resilient supply chains. More work could be done to adequately capture the returns to resilience, balanced with the efficiency trade-offs, to better inform public discourse and help prepare for future events that would be disruptive to the different stages of food production.

4.2. Empirical Methods in Supply Chain Resilience

The food systems literature has grappled with how to define and measure resilience within a food system, and whether resilience itself is the intended outcome or an intermediate step (for reviews, see Urruty et al. 2016, Ansah et al. 2019, Béné 2020, Fan et al. 2021, Vos et al. 2022b). Whether resilience is seen as a means to an end (resilience capacity) versus an ultimate outcome determines how it is measured (Ansah et al. 2019). When viewed as an ultimate outcome, food system resilience is often measured with household indicators of food security. Increasingly, however, resilience is measured as a capacity, a means to achieve food security, using a combination of proxy variables, indicators, and household survey data. Challenges exist in establishing a causal relationship between resilience and household food security due to potential endogeneity problems. These include the potential for interdependence between the resilience attributes (absorptive capacity, adaptive capacity, and transformative capacity) and the potential for a reverse causal relationship between household food security and resilience capacity (for a detailed discussion, see Ansah et al. 2019).

A primary challenge for economists studying supply chain resilience at a sectoral level is a lack of historical data to provide insight into the performance of supply chains under extreme volatility. Several studies use industry-level data to assess the resiliency of food supply chains to

recent disruptive events broadly. Ramsey et al. (2021), for example, use average prices from the US meat industry to map price transmission along the supply chain using error correction models.⁴ These types of ex post analyses of specific shocks are indeed valuable and relevant lessons in food supply chain resilience. However, future disruptions may elicit unique vulnerabilities in the food industry, and the impacts of past disruptions may not necessarily be indicative of future resilience. It is difficult to draw convincing conclusions from industry-wide studies about whether different structures or strategies might make food supply chains more resilient.

Many modern empirical applications in economics are defined by using granular data sets and exploiting plausibly exogenous variation in treatment variables. In the context of supply chain resilience, both dimensions are difficult to achieve in practice. First, detailed data on production processes and prices at the intermediary and retail level are typically proprietary and not available to researchers. Cooper et al. (2023) and Padilla et al. (2023) are valuable exceptions to this, utilizing plant-level data available through the US Department of Agriculture. Second, even when detailed data exist, fleshing out credible counterfactual outcomes is rarely possible for major disruptions, such as the COVID-19 pandemic. Systemic shocks impact almost all agents simultaneously, leaving no viable reference point or control group to reasonably compare across sectors or industries. For these reasons, dialog around supply chain resilience has largely been descriptive and conceptual in nature. Several recent exceptions exist, however, and can provide some insight into useful paths forward.

As noted in Section 2, plant disease and foodborne illness are two sources of vulnerabilities in modern food production. Spalding et al. (2023) document the economic consequences of *E. coli* outbreaks for all agents along the romaine lettuce supply chain. This is an example of researchers partnering with private industry—in this case, a large lettuce processor—to obtain proprietary data and decompose the distributional effects of outbreaks for the industry. Such partnerships can provide new insights into the economic implications of and responses to unexpected supply chain shocks.

Survey data may also provide novel insights beyond industry-level data. Stevens & Teal (2024) conduct a survey of agribusinesses following the COVID-19 pandemic to assess industry response in agrifood sectors. They study how vertical and horizontal diversification among US firms in the agrifood supply chain impacted performance following the onset of COVID-19. To best compare firms with varying degrees of diversification, they construct weighted-matching estimators based on observable firm characteristics. They then quantify the extent to which COVID-19 impacted firms' operations based on their degree of horizontal and vertical differentiation.

Lastly, Gnutzmann et al. (2020) leverage a nuanced source of variation in the potash market. They study the resilience of the potash mining industry by assessing how shocks permeate to mining companies with two different business structures, syndicates (or cartel-like agreements), and the competitive fringe. They utilize the incidences of mining collapse as shocks to the capacity of supply. They study the resulting equilibrium price and quantity impacts for potash in the months following these disasters. They also compare the impacts in a related fertilizer market, phosphorus, as a control comparison. They conclude, based on the lack of variability in prices and quantities following collapses in syndicate mines, that the structure of potash syndicates provides a degree of resilience in equilibrium outcomes.

Even if settings do provide a feasible pathway to uncover causal impacts on resilience, data constraints may pose an issue. The data collection process itself may also be impacted by the disruption and may not fully reflect market conditions. Stockouts in grocery stores are one example

⁴For more discussion on the price transmission literature in agricultural economics, see von Cramon-Taubadel & Goodwin (2021).

of this. Where stockouts occurred, scanner data may be censored by the amount of stock available and may not reflect willingness to pay and demand for these food products. Using detailed scanner data, McLaughlin et al. (2023) make progress on this front by introducing an algorithmic method to identify the rate at which stockouts happened. As stockouts are a common phenomenon following extreme events, accurately identifying stockouts and where they occur can provide further insight into consumer welfare.

As empirical work on supply chain resilience advances, it is imperative that the discussion moves beyond the generalized impacts of events, such as COVID-19, and more toward the mechanisms that drive these impacts. Further evidence on the latter can better identify the magnitude of future threats and the potential alternatives to mitigate the impacts. To this end, distributional analyses of food supply chains can greatly advance knowledge about the resilience of food supply chains. This can take the form of both distributional analysis of impacts on individuals (e.g., food insecure or those in low- and middle-income countries and food deserts) and the distribution of economic outcomes over time as shocks occur. Empirical models that test second- (variance) and third- (skewness) ordered moments could be adapted to be useful in measuring the resilience of food supply chains. Antle (1983) provides the framework for this analysis, but to date, it has largely been used to assess on-farm production and household resilience.⁵ Paired with appropriate data, this framework could extend to other stages of the food supply chain and test how different characteristics (e.g., differentiation, concentration, excess capacity) impact higher-ordered moments of supply chain performance measures.

5. CONCLUDING REMARKS

Designing effective policies to enhance food supply chain resilience requires a clear understanding of what we mean by resilience, how we define and measure resilience, and the intended outcomes of enhanced resilience. Focusing purely on supply chain resilience as an end policy goal in and of itself misses the point. Instead, the implications of enhanced supply chain resilience for a broader set of welfare and distributional outcomes—identifying winners, losers, and trade-offs—is a more useful endeavor and one in which economic analysis plays a central role.

The concept of supply chain resilience is one with varied definitions and interpretations depending on the perspective. This article seeks to harmonize the various concepts, summarize knowledge to date about food supply chain resilience, and offer some considerations for economic research on the topic. Substantial work has revealed the impacts of the COVID-19 pandemic and other events on agricultural production and food manufacturing outcomes and has underscored vulnerabilities (and strengths) of modern food supply chains. Yet, considering the potential for future disruptive events, there remains substantial room for economic research to help inform policy to prepare for the myriad of potential threats to food production and distribution.

Future research could explore supply chain resilience and welfare outcomes in the context of different types of exogenous shocks. Widespread, correlated shocks may have a greater potential to cause disruption relative to isolated, idiosyncratic shocks but may also have a lower probability of occurring. There is scope for work modeling the effect of different types of shocks on resilience outcomes and the extent to which policy initiatives to build robustness and enhance adaptability are effective under different types of shocks.

The emerging body of literature in the wake of the COVID-19 pandemic suggests that commodity sectors and supply chain segments differ in their vulnerability due to physical attributes (storability versus perishability), reliance on labor (degree of mechanization), dependence on

⁵ Indeed, Cisse & Barrett (2018) apply this approach to construct a measure of development resilience.

lateral supply chains (logistics, transportation networks), and the existence of alternative sources of supply. Analysis that further unpacks the nature and determinants of these vulnerabilities within and across commodity sectors and the extent to which they were specific to the unique circumstances of the COVID-19 pandemic or signal a systemic vulnerability would be timely.

Robust empirical analysis of supply chain resilience requires good data and an appropriate counterfactual. Ex ante analysis of potential vulnerabilities to future shocks requires detailed simulation analyses built with a thorough understanding of the structure of the agrifood sector, the nature of both vertical and lateral supply chain relationships within the sector, and the scope of cross-sectoral and regional linkages. Special attention to the outcome variables of interest is warranted, including distributional and welfare outcomes. Ex post analysis to identify, isolate, and measure the impact of a given shock on supply chain resilience for a firm or for an agrifood sector requires detailed firm- or industry-level data that are often proprietary in nature or difficult to obtain. Cooperation with public sector agencies with access to high-quality firm-level (or even transaction-level) data will be increasingly important for policy-relevant research going forward. If we want to better understand the effect of climate, geopolitical, or disease shocks on supply chain resilience, the public good aspect of timely access to such data needs to be recognized.

Further research is needed to address the extent to which private market incentives to invest in resilience strategies (at the firm or industry level) reflect a socially optimal balance between efficiency and resilience. The degree to which firm-level resilience tactics deliver resilience benefits beyond the firm itself (to consumers or other supply chain actors) warrants closer attention. If not, then policy interventions may be needed to achieve broader resilience goals. This requires a better understanding of how to model and measure the public benefits of supply chain resilience and the effectiveness of policy interventions to enhance resilience. A central consideration in examining resilience strategies, both at the firm level and with respect to policy interventions, is evaluating the net benefits of building enhanced resilience to extreme or infrequent events given the likelihood that this introduces inefficiencies during normal times. In evaluating the resiliency-efficiency trade-off, further work could examine how (and whether) to assign a premium to enhanced resilience to infrequent, yet extreme, events.

The world is unpredictable, and the agrifood sector grapples with unpredictability on a daily basis, from extreme weather, to transportation bottlenecks, and to political instability in domestic and international markets. As food supply chains continue to be buffeted by geopolitical, climate, and plant and animal disease-related shocks, the topic of supply chain resilience will continue to resonate in the public consciousness. There is scope for deepening our understanding of food supply chain resilience, from the perspective of both firm-level strategies to enhance resilience and industry-level strategies to enhance sectoral resilience. From a policy perspective, understanding where, when, and even whether policy interventions are needed to enhance resilience depends on carefully designed economic analysis of the inherent trade-offs embedded in a policy intervention and the relevant welfare outcomes. This remains a rich and exciting area for applied economics research.

SUMMARY POINTS

1. Food supply chains experienced widespread disruption in recent years and remain vulnerable to future extreme events.
2. Resilience is a multidimensional concept, examined through various disciplinary lenses and at different levels of analysis (food system, household, firm, sector).

3. It is critical that researchers clearly state their definition and measure of resilience to harmonize lessons across studies and subject areas.
4. Current theoretical and empirical methodologies may not fully capture the complex dimensions of resilience, the societal value of resilient food supply chains, or the extent of trade-offs involved with suggested interventions.
5. A large literature has documented the impacts of COVID-19 on food supply chains, but more work is needed to prepare and inform policy to build future food supply chain resilience.

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